2022 WASSCE CORE MATHEMATICS 2 SOLUTION

WASSCE FOR SCHOOL CANDIDATES, 2022
GENERAL MATHEMATICS / MATHEMATICS (CORE) 2
(GHANA)

QUESTION	DETAILS	MARKS
1. (a)	$ar^7 = \frac{1}{256}$ ....(1) $ar^2 = 4$.....................(2)	B1 for either (1) or (2)
	$\frac{ar^7}{ar^2} = \frac{1}{256} \div 4$	M1 for dividing
	$r^5 = \frac{1}{1024}$ $r^5 = 4^{-5}$ $r^5 = \frac{1}{4^5}$ $r^5 = \left(\frac{1}{4}\right)^5$	M1 for simplifying
	$r = \frac{1}{4}$	A1 for $r = \frac{1}{4}$ accept
(b)	$a\left(\frac{1}{4}\right)^2 = 4$	M1 for substituting
	$\frac{a}{16} = 4$ $a = 64$	A1 for $a = 64$
(c)	$2nd\ term = ar = 64 \times \frac{1}{4} = 16$	M1 for substituting
	$sum = 64 + 16$ $= 80$	A1 for 80
		[8 marks]

QUESTION	DETAILS	MARKS
2. (a)	$12\frac{1}{2}\%\ of\ votes = 275$ $100\% = \frac{100\times 275}{12\frac{1}{2}}$ $= 2{,}200$ Votes obtained by winner $= \frac{87\frac{1}{2}}{100}\times 2{,}200$ $= 1{,}925$ or $2{,}200 - 275 = 1{,}925$	**B1 for** $12\frac{1}{2}\%\ of\ votes = 275$ **or equivalent** **M1 for** $\frac{100\times 275}{12\frac{1}{2}}$ **or equivalent** **A1 for 2,200** **M1 for** $\frac{87\frac{1}{2}}{100}\times 2{,}200$ **or equivalent** **A1 for 1,925**
(b)	55% of votes $= 2{,}200$ Total number of eligible voters $= \frac{100\times 2{,}200}{55}$ $= 4{,}000$	**B1 for 55% of votes = 2,200 or equivalent** **M1 for** $\frac{100\times 2{,}200}{55}$ **or equivalent** **A1for 4,000** **[8 Marks]**

QUESTION	DETAILS	MARKS
3. (a)	volume of tank $= 1.2 \times 1.2 \times 1.2$ $= 1.728m^3$ 1.728×1000 $= 1{,}728\,litres$	**M1 for finding volume** **A1 for 1.728m³** **M1 for conversion** **A1 for 1,728 litres**
(b)	Number of containers $= \frac{1{,}728}{54}$ $= 32$	**M1 for dividing** **A1 for 32**
(c)	Amount paid $= 32 \times 1.75$ $=$ GH₵56.00	**M1 for multiplying** **A1 for GH₵56.00** **(-1 ou/wu once only)** **[8 Marks]**

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QUESTION	DETAILS	MARKS
4.	$x = 7m - 16$ ....(1) Either of these	**M1 for either (1) or (2)**
	$y = 180 - x$ ....(2)	
	$y = 180 - (7m - 16)$ solving	**M1 for solving**
	$= 180 - 7m + 16$	
	$y = 196 - 7m$	**A1 for $y = 196 - 7m$ or equivalent**
	$\frac{x}{y} = \frac{2}{7}$ either	**B1 for ratio**
	$x = \frac{2}{7}y$	
	$7m - 16 = \frac{2}{7}y$	**M1 for equating**
	$7m - 16 = \frac{2}{7}(196 - 7m)$	**M1 for substituting**
	$49m - 112 = 392 - 14m$	
	$63m = 504$	**M1 for solving**
	$m = 8$	**A1 for m = 8**
		[8 Marks]

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QUESTION	DETAILS	MARKS
5. (a)	$\cos y = \frac{12}{13}, \tan y = \frac{5}{12}$	**B1 for either $\left(cosy = \frac{12}{13}, tany\frac{5}{12}\right)$**
	$\frac{3\cos y - 2\sin y}{6\tan y} = \frac{3\left(\frac{12}{13}\right) - 2\left(\frac{5}{13}\right)}{6\left(\frac{5}{12}\right)}$ M1 all correct	**M1 for substituting**
	Simplification	**M1 for simplifying**
	$= \frac{4}{5}$	**A1 for $\frac{4}{5}$**
(b)	$N = \{10, 11, 12, .., 28, 29, 30\}$	**B1 for listing the numbers** n(N)
	Multiples of $3 = \{12, 15, 18, 21, 24, 27, 30\}$ Multiples of $4 = \{12, 16, 20, 24, 28\}$ Multiples of 3 and $4 = \{12, 24\}$	**B1 for listing either of the sets or stating the no. of elements in either of the sets**
	$P(\text{multiples of 3 or 4}) = \frac{7}{21} + \frac{5}{21} - \frac{2}{21}$ all 3 items correct	**M1 for finding probability**
	$= \frac{10}{21}$	**A1 for $\frac{10}{21}$**
		[8 Marks]

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QUESTION		DETAILS	MARKS
6.	(a)	$2cm\ to\ 1\ unit\ on\ x-axis$	**B1 for x – axis**
		$2cm\ to\ 2\ units\ on\ y-axis$	**B1 for y – axis**
	(b)	$x=1, x=5$	**M1 for forming equation**
		$y=(x-1)(x-5)$	**A1 for x^2-6x+5**
		$=x^2-6x+5$	
		$a=1$	**A1 for $a=1$**
		$b=-6$	**A1 for $b=-6$**
		$c=5$	**A1 for $c=5$**
(c)		$x=-0.3\pm0.1$ any of the correct	**M1 for reading**
		$x=6.3\pm0.1$	**A1 for $x=-0.3\pm0.1$**
			A1 for $x=-6.3\pm0.1$
	(d)	$(3,-4)$ accept only coordinate form	**B1 for (3, - 4)**
	(e)	$1<x<5$	**B1 for $1<x<5$**
		$\{1<x<5\}$	**[12 Marks]**

QUESTION	DETAILS	MARKS
7.(a)	Total ratio = 5 + 12 + 3 = 20 No marks	
	Labour = $\frac{5}{20} \times 120{,}000$	**M1 for any ratio**
	= GH₵30,000.00	**A1 for GH₵30, 000. 00**
	Material = $\frac{12}{20} \times 120{,}000 = GH₵72{,}000.00$	**A1 for GH₵72, 000. 00**
	Consultancy = $\frac{3}{20} \times 120{,}000$ = GH₵18,000.00	**A1 for GH₵18, 000. 00**
	Increase in labour = $\frac{2y}{100} \times 30{,}000 = 600y$……..(1)	**B1 for either $\frac{2y}{100}$ or $\frac{2.5y}{100}$**
	Increase in materials = $\frac{2.5y}{100} \times 72{,}000 = 1{,}800y$…….(2)	**M1 for either (1) or (2)**
	$30{,}000 + 600y = \frac{2}{5}(72{,}000 + 1{,}800y)$ solving	**M1 for equation**
	$150{,}000 + 3{,}000y = 144{,}000 + 3{,}600y$ solving	**M1 for simplifying**
	$600y = 6{,}000$	
	$y = 10$	**A1 for $y = 10$**
(b)	New cost of building	correct substitution only
	= 18,000+(30,000+(600 ×10)) + (72,000 + (1800 × 10))	**M1 for substituting**
	= 18,000 + 36,000 + 90, 000 Check follow through	**M1 for simplifying**
	= GH₵144,000.00	**A1 for GH₵144,000.00**
	-1 roll off	once only **[12 Marks]**

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QUESTION	DETAILS	MARKS
8. (a)	In 1 hour Kojo can do $\frac{1}{5}$ $x/5$ any one	**B1 for $\frac{1}{5}$ or $\frac{1}{4}$**
	In 1 hour Ama can do $\frac{1}{4}$ $x/4$	
	In 1 hour both can do	
	$\frac{1}{x} = \frac{1}{5} + \frac{1}{4}$	**M1 for equation**
	$4x + 5x = 20$	**M1 for clearing fractions**
	$9x = 20$ solving	**M1 for solving**
	$x = \frac{20}{9} = 2\frac{2}{9}$hours	**A1for $2\frac{2}{9}$hours** $\frac{20}{9}$ hrs **(Accept 2.22 hours)**
(b)	$h = \sqrt{6^2 - 3^2}$	**M1 for use of Pythagoras theorem**
	$= \sqrt{27}$	
	$= 5.196\text{cm or } 3\sqrt{3}cm$	**A1 for 5.196 cm or $3\sqrt{3}cm$**
	Area $= \frac{1}{2} \times 6 \times 5.196$	**M1 for finding area**
	$= 15.588cm^2 \text{ or } 9\sqrt{3}\text{cm}^2$	**A1 for 15.588 cm² or $9\sqrt{3}$ cm²**
	Area of rectangular face $= 12 \times 6$	
	$= 72cm^2$	**B1 for 72 cm²**
	Total surface area $= (2 \times 15.588) + (3 \times 72)$	**M1 for finding total surface area**
	$= 247.172$	
	$= 247.2cm^2$	247.2cm² **A1 for ~~242~~.2cm²** **(-1 ou/wu once only)** **[12 Marks]**

QUESTION	DETAILS	MARKS

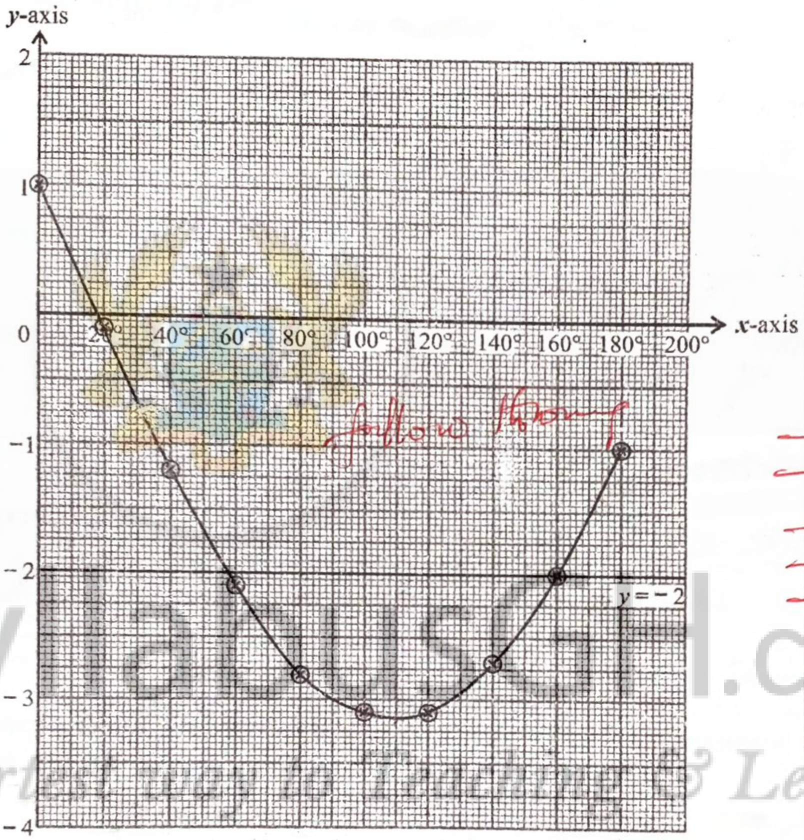
9 (a)

x	0^0	20^0	40^0	60^0	80^0	100^0	120^0	140^0	160^0	180^0
y	1.0	-0.1	-1.2	-2.1	-2.8	-3.1	-3.1	-2.7	-2.0	-1.0

B3$\left(-\frac{1}{2}ee\right)$

B2 for non-use of table

(b)



follow through

B3$\left(-\frac{1}{2}ee\right)$

- non-labelling
- wrong plotting
- wrong scale
- use of ruler
- non-joining of points

(c)(i) $y = -2$ — B1 for $y = -2$

must evidence be shown on graph

$\{x : x = 58^0 \pm 2^0, 160^0 \pm 2^0\}$ — M1 for reading one value correct; A1 for all correct

reading one value correct

(ii) $110^0 < x \le 180^0$ — M1A1

(108 – 112, 14.5 — 15.5)

(iii) Minimum point $(110^0 \pm 2^0, -3.15 \pm 0.05)$ — B1 for minimum point

all in bracket

[12 marks]

QUESTION	DETAILS	MARKS
10. (a)	$2x + 3x + (4x - 1) + x + (x - 2) + (x - 3) = 42$ Equation $=42$	**M1 for adding** **A1 for equating to 42**
	$12x = 48$ so Pump	**M1 for solving**
	$x = 4$	**A1 for $x = 4$**
(b)	Subt of any 2 $\text{Mean} = \dfrac{(7\times 8)+(8\times 12)+(9\times 15)+(10\times 4)+(11\times 2)+(12\times 1)}{42}$	**M1 for any two products correct**
	$\text{Mean} = \dfrac{56+96+135+40+22+12}{42}$	**A2 $\left(\frac{-1}{2} ee\right)$ for all correct**
	$= \dfrac{361}{42} = 8.595$	M1 ~~**B1**~~ **for 361** $\frac{}{42}$
	$= 9$ *years*	**A1 for 9 years**
(c)	Children not less than 9 years $= 15 + 4 + 2 + 1$ only $= 22$	**B1 for 22**
	$P(\text{not less than } 9) = \dfrac{22}{42}$	**M1 for finding prob.**
	$= \dfrac{11}{21}$ or 0.52	**A1 for $\frac{11}{21}$**
		[12 Marks]

QUESTION	DETAILS	MARKS
11(a)	$\cos(2x+15)=\cos[90-(x-30)]$ M1A1	**M1A1**
	$\cos(2x+15)=\cos(90-x+30)$	
	$2x+x=120-15$	**M1 for solving**
	$\cancel{3x+x=120-15}$	
	$3x=105$	
	$x=35^0$	**A1 for 35^0**
(b)(i)	$\angle CBD=\angle FCE$	
	$180-(2x+34)=48$	**M1 for equation**
	$180-2x-34=48$	
	$146-2x=48$	
	$98=2x$	**M1 for solving**
	$x=49^0$	**A1 for 49^0 (cao)**
(ii)	$\angle BCD+\angle BDC+\angle CBD=180$	
	$\frac{1}{3}\angle BDC+\angle BDC+48=180$ at least 2 correct	**M1A1 for equation**
	$\frac{4}{3}\angle BDC=180-48$	
	$\frac{4}{3}\angle BDC=132$ Solving	**M1 for solving**
	$\angle BDC=\frac{132\times 3}{4}$	**M1 for simplifying**
	$=99^0$	**A1 for 99^0**
	-1 OR [illegible]	
		[12 Marks]

QUESTION	DETAILS	MARKS
12. (a)(i)	Prime numbers = {2, 3, 5, 7, 11, 13} at least 4 prime nos correct	B1 for prime numbers
	$P\{\text{Prime}\} = \frac{6}{15}$	M1 for $\frac{6}{15}$
	$= \frac{2}{5}$ accept 0.4	A1 for $\frac{2}{5}$
(ii)	Multiples of 3 = {3, 6, 9, 12, 15}	
	$P(\text{multiples of } 3) = \frac{5}{15}$ 5	M1 for $\frac{5}{15}$
	$= \frac{1}{3}$ 0.33	A1 for $\frac{1}{3}$
(iii)	Numbers divisible by 5 = {5, 10, 15}	
	$P(\text{nos. divisible by } 5) = \frac{3}{15}$	M1 for $\frac{3}{15}$
	$= \frac{1}{5}$	A1 for $\frac{1}{5}$
(b)	$\overrightarrow{PQ} = 4\overrightarrow{SR}$	
	$\overrightarrow{OQ} - \overrightarrow{OP} = 4\left(\overrightarrow{OR} - \overrightarrow{OS}\right)$ Correct Substitution sly	
	$\overrightarrow{OQ} - \begin{pmatrix}4\\4\end{pmatrix} = 4\left[\begin{pmatrix}12\\8\end{pmatrix} - \begin{pmatrix}8\\8\end{pmatrix}\right]$	M1 for substituting
	$\overrightarrow{OQ} - \begin{pmatrix}4\\4\end{pmatrix} = 4\begin{pmatrix}4\\0\end{pmatrix}$ Simplifying	M1 for simplifying
	$\overrightarrow{OQ} - \begin{pmatrix}4\\4\end{pmatrix} = \begin{pmatrix}16\\0\end{pmatrix}$	M1 for solving
	$\overrightarrow{OQ} = \begin{pmatrix}16\\0\end{pmatrix} + \begin{pmatrix}4\\4\end{pmatrix}$ Solving	
	$= \begin{pmatrix}20\\4\end{pmatrix}$	A1 for $\begin{pmatrix}20\\4\end{pmatrix}$
	$Q = (20, 4)$	A1 for $Q = (20, 4)$
		[12 Marks]

QUESTION	DETAILS	MARKS
13 (a)	17:45 – 13:15 = 4 hours 30 min.	B1 for 4 hrs 30 min.
	Angle gained = (4× 30) + 15	M1 for finding angle
	$= 135^0$	A1 for 135^0
	Area swept $= \frac{135}{360} \times \frac{22}{7} \times 3.5^2$	M1 for finding area
	= 14.4375	M1 for simplifying
	= 14.4 cm^2	A1 for 14.4 cm^2 (-1ou/wu)
(b)	$\sqrt{(6-3)^2 + (x+4)^2} = 3\sqrt{10}$	M1 for finding distance
	$\sqrt{9 + (x+4)^2} = 3\sqrt{10}$	
	$9 + (x+4)^2 = 9 \times 10$	M1 for simplifying
	$9 + x^2 + 8x + 16 = 90$	
	$x^2 + 8x + 25 - 90 = 0$	
	$x^2 + 8x - 65 = 0$	A1 for quadratic equation
	$x(x+13) - 5(x+13) = 0$ one factor correct	
	$(x-5)(x+13) = 0$	M1 for factorizing (one factor correct)
	$x = 5$	A1 for $x = 5$
	$x = -13$	A1 for $x = -13$
		[12 marks]